Chapter 4 Multivariate Normal Population Statistical Inference

Content

§ 4.4 Single Population Mean Component Structural Relationship Test

- Suppose $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n$ is a sample of multivariate normal population $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma}>0$, $n>p$. Hypothesis Test:

$$H_0: \boldsymbol{C\mu}=\boldsymbol{\varphi}, \quad H_1: \boldsymbol{C\mu}\neq\boldsymbol{\varphi} \tag{4.4.2}$$

where $\boldsymbol{C}$ is a known $k \times p$ matrix, $k < p$, rank$(\boldsymbol{C})=k$ and $\boldsymbol{\varphi}$ is a given k-vector.

On the basis of multivariate normal distribution properties,

$$\boldsymbol{Cx} \sim N_k(\boldsymbol{C\mu}, \boldsymbol{C\Sigma C'})$$

Since

$$\operatorname{rank}(\boldsymbol{C\Sigma C'}) = \operatorname{rank}\left[\boldsymbol{C\Sigma}^{\frac{1}{2}}\left(\boldsymbol{C\Sigma}^{\frac{1}{2}}\right)'\right] = \operatorname{rank}\left(\boldsymbol{C\Sigma}^{\frac{1}{2}}\right) = \operatorname{rank}(\boldsymbol{C}) = k$$

Then $\boldsymbol{C\Sigma C'}>0$. We can construct Hotelling statistic to test (4.4.2), which is given by

$$T^2 = n\left(\boldsymbol{C\bar{x}}-\boldsymbol{\varphi}\right)'\left(\boldsymbol{CSC'}\right)^{-1}\left(\boldsymbol{C\bar{x}}-\boldsymbol{\varphi}\right),$$

where $T^2 \sim T^2(k, n-1)$. When $H_0:\ \boldsymbol{C\mu}=\boldsymbol{\varphi}$ is true,

$$\frac{n-k}{k(n-1)}T^2 \sim F(k, n-k)$$

- **Properties**:

$$\frac{n-p+1}{pn}T^2(p,n) \sim F(p, n-p+1)$$

where n is the free degree of the above Wishart Distribu

For a given significance level α, the rejection rule is:

If $T^2 \geq T_\alpha^2$, reject H_0.

where $T_\alpha^2 = \dfrac{k(n-1)}{n-k}F_\alpha(k, n-k)$.

- Especially, when the following hypothesis will be tested,

$$H_0:\ \boldsymbol{C\mu}=\boldsymbol{0},\quad H_1:\ \boldsymbol{C\mu}\neq\boldsymbol{0}$$

then T^2 can be simplified as

$$T^2 = n\boldsymbol{\bar{x}}'\boldsymbol{C}'\left(\boldsymbol{CSC'}\right)^{-1}\boldsymbol{C\bar{x}}$$

This leads to the following profile analysis(轮廓分析)

4.4 Contour analysis （轮廓分析）

- Example 4.3.1 Suppose $\boldsymbol{x}\sim N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$， $\boldsymbol{\mu}=(\mu_1,\mu_2,\cdots,\mu_p)'$， $\boldsymbol{\Sigma}>0$， $\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_n$ is a sample chosen from the above population. Hypothesis test is

$$H_0:\ \mu_1=\mu_2=\cdots=\mu_p,\quad H_1:\ \mu_i\neq\mu_j, \text{ at least one couple } i\neq j,$$

where H_0 implies the profile is horizontal. Let

$$\boldsymbol{C}=\begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ 1 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & -1 \end{pmatrix}$$

then the above hypothesis is expressed as:

$$H_0:\ \boldsymbol{C\mu}=\boldsymbol{0},\quad H_1:\ \boldsymbol{C\mu}\neq\boldsymbol{0}.$$

The test statistic is

$$T^2=n\bar{\boldsymbol{x}}'\boldsymbol{C}'\left(\boldsymbol{CSC}'\right)^{-1}\boldsymbol{C}\bar{\boldsymbol{x}},$$

where $T^2\sim T^2(p-1,n-1)$.

§ 4.4 轮廓分析

设对同一个单元(个人、商店、小块土地等)施加 p 种处理(如测验,问卷调查等)或在相继 p 个时间段内重复测量,依次得到测量值 $x_1, x_2, \cdots, x_p$,其相应的均值依次为 $\mu_1, \mu_2, \cdots, \mu_p$。

图 4.4.1 是将点 $(1, \mu_1), (2, \mu_2), \cdots, (p, \mu_p)$ 用直线连接起来的折线图,称为总体的轮廓(profile)。轮廓分析是轮廓的分析或多个轮廓的比较,它常用于分析比较相继一连串的心理测试或其他测试中。我们这里讨论单总体(或单组)和两总体(或两组)的轮廓分析。

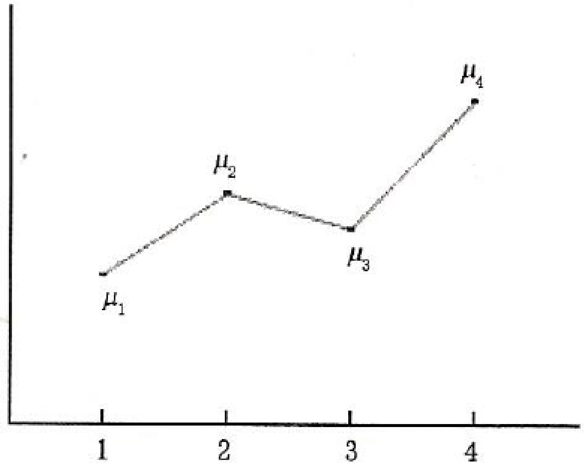

图 4.4.1 总体轮廓($p=4$)

一、单总体的轮廓分析

为了比较均值 $\mu_1, \mu_2, \cdots, \mu_p$,一个基本的假设为轮廓是水平的,即

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_p, \quad H_1: \mu_i \neq \mu_j, \text{至少存在一对 } i \neq j \tag{4.4.1}$$

For a given significance level α, the rejection rule is :

$$\text{If } T^2 \geq T_\alpha^2, \quad \text{reject } H_0$$

where $T^2 \sim T^2(p-1, n-1)$ and

$$T_\alpha^2 = \frac{(p-1)(n-1)}{n-p+1} F_\alpha(p-1, n-p+1)$$

Properties :

$$\frac{n-p+1}{pn} T^2(p,n) \sim F(p, n-p+1)$$

where n is the free degree of the above Wishart Distribu

- Since $\boldsymbol{C}$ is row full rank and every row is a comparison vector (which have one "1" and one "−1" elements and others are 0) , $\boldsymbol{C}$ is called comparison matrix(对比矩阵) .
- In this example, comparison matrix $\boldsymbol{C}$ is not unique, like

$$\boldsymbol{C}^* = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ 0 & 1 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{pmatrix}$$

- Example 4.3.2. In Example 4.2.1, assume human has such a universal law that the average dimensional proportion of high and chest circumference and upper arm girth is 6:4:1. We hope the data of Table 4.2.1 conform to the above regulation. Then we need to test

$H_0:\ \mu_1/6=\mu_2/4=\mu_3,\quad H_1:\ \mu_1/6, \mu_2/4, \mu_3$ at least one couple are not equal

Let

$$\boldsymbol{C}=\begin{pmatrix} 2 & -3 & 0 \\ 1 & 0 & -6 \end{pmatrix}$$

$$\boldsymbol{S}=\begin{pmatrix} 31.600 & 8.040 & 0.500 \\ 8.040 & 3.172 & 1.310 \\ 0.500 & 1.310 & 1.900 \end{pmatrix}$$

Then the above hypothesis can be expressed as

$$H_0:\ \boldsymbol{C\mu}=\boldsymbol{0},\quad H_1:\ \boldsymbol{C\mu}\neq\boldsymbol{0}$$

After calculation,

$$\boldsymbol{C\bar{x}}=\begin{pmatrix} -16.6 \\ -4.0 \end{pmatrix},\quad \boldsymbol{CSC'}=\begin{pmatrix} 58.468 & 56.660 \\ 56.660 & 94.000 \end{pmatrix}$$

Therefore,

$$(\boldsymbol{CSC'})^{-1}=(2285.6364)^{-1}\begin{pmatrix} 94.000 & -56.660 \\ -56.660 & 58.468 \end{pmatrix}$$

So

$$T^2 = n\bar{\boldsymbol{x}}'\boldsymbol{C}'\left(\boldsymbol{CSC}'\right)^{-1}\boldsymbol{C}\bar{\boldsymbol{x}} = 6\times 8.450 = 50.700$$

Since the above $T^2 \sim T^2(k, n-k)$, we have

$$T^2_{0.01} = \frac{k(n-1)}{n-k}F_{0.01}(k, n-k) = \frac{2\times 5}{4}\times 18.0 = 45 < T^2$$

we reject H_0, namely this group of data is not consistent to human universal law (p=0.008).

- The above $\boldsymbol{C}$ can be chosen as

$$\boldsymbol{C}^* = \begin{pmatrix} 0 & 1 & -4 \\ 1 & 0 & -6 \end{pmatrix}$$

and the result remains unchanged.

§ 4.5 Two Population Mean Component Structural Relationship Test

- Suppose two independent sample $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_{n_1}$ and $\boldsymbol{y}_1, \boldsymbol{y}_2, \cdots, \boldsymbol{y}_{n_2}$ are respectively chosen from $N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma}>0$, $n_1+n_2-2 \geq p$. Test

$$H_0: \ \boldsymbol{C}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2)=\boldsymbol{\varphi}, \quad H_1: \ \boldsymbol{C}(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2)\neq\boldsymbol{\varphi}$$

where $\boldsymbol{C}$ is a known $k \times p$ matrix, $k<p$, rank($\boldsymbol{C}$)=k and $\boldsymbol{\varphi}$ is a given k-vector. Test statistic is constructed as

$$T^2 = \frac{n_1 n_2}{n_1+n_2}\left[\boldsymbol{C}\left(\bar{\boldsymbol{x}}-\bar{\boldsymbol{y}}\right)-\boldsymbol{\varphi}\right]'\left(\boldsymbol{C}\boldsymbol{S}_p\boldsymbol{C}'\right)^{-1}\left[\boldsymbol{C}\left(\bar{\boldsymbol{x}}-\bar{\boldsymbol{y}}\right)-\boldsymbol{\varphi}\right]$$

where $\boldsymbol{S}_p$ is the joint unbiased estimate of $\boldsymbol{\Sigma}$. When H_0 is true,

$T^2 \sim T^2(k, n_1+n_2-2)$. It follows that

$$\frac{\left(n_1+n_2-k-1\right)T^2}{k\left(n_1+n_2-2\right)} \sim F\left(k, n_1+n_2-k-1\right)$$

Properties:

$$\frac{n-p+1}{pn} T^2(p,n) \sim F(p, n-p+1)$$

where n is the free degree of the above Wishart Distribu

Rejection rule: If $T^2 \geq T_\alpha^2$, reject H_0, where

$$T_\alpha^2 = \frac{k\left(n_1+n_2-2\right)}{n_1+n_2-k-1} F_\alpha\left(k, n_1+n_2-k-1\right)$$

- Example 4.5.1 A certain kind of product has two brands, A and B. Randomly chose 5 samples respectively from a group products of rand A and B and measure the same 5 indexes. Data are listed in Table 4.5.1. On the condition of multivariate normal hypothesis, try to figure out whether there exist a significant difference in the indices between Brand A and B, which is corresponding to the contour analysis of two populations. The question can be translated into the following test:

$$H_0: \boldsymbol{C}(\boldsymbol{\mu}_{\mathrm{A}}-\boldsymbol{\mu}_{\mathrm{B}})=\mathbf{0}, \quad H_1: \boldsymbol{C}(\boldsymbol{\mu}_{\mathrm{B}}-\boldsymbol{\mu}_{\mathrm{A}})\neq\mathbf{0}$$

where

$$\boldsymbol{C} = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

Table 4.5.1 Index Value of Brand A and Brand B

Sample \ Index		1	2	3	4	5
A	1	11	18	15	18	15
	2	33	27	31	21	17
	3	20	28	27	23	19
	4	18	26	18	18	9
	5	22	23	22	16	10
Mean		20.8	24.4	22.6	19.2	14.0
B	1	18	17	20	18	18
	2	31	24	31	26	20
	3	14	16	17	20	17
	4	25	24	31	26	18
	5	36	28	24	26	29
Mean		24.8	21.8	24.6	23.2	20.4

Test statistic

$$T^2 = \frac{n_1 n_2}{n_1 + n_2}\left(\boldsymbol{C}(\overline{\boldsymbol{x}}_B - \overline{\boldsymbol{x}}_A)\right)'\left(\boldsymbol{C}\boldsymbol{S}_p\boldsymbol{C}'\right)^{-1}\boldsymbol{C}\left(\overline{\boldsymbol{x}}_B - \overline{\boldsymbol{x}}_A\right)$$

After calculation

$$\overline{\boldsymbol{x}}_B - \overline{\boldsymbol{x}}_A = \left(4.0, -2.6, 2.0, 4.0, 6.4\right)'$$

$$\boldsymbol{S}_p = \begin{pmatrix} 72.700 & 33.025 & 41.650 & 18.675 & 22.300 \\ 33.025 & 21.250 & 21.300 & 12.725 & 11.925 \\ 41.650 & 21.300 & 41.300 & 16.350 & 9.850 \\ 18.675 & 12.725 & 16.350 & 11.450 & 10.200 \\ 22.300 & 11.925 & 9.850 & 10.200 & 21.650 \end{pmatrix}$$

$$C\left(\bar{\boldsymbol{x}}_B - \bar{\boldsymbol{x}}_A\right) = \left(6.6, -4.6, -2.0, -2.4\right)'$$

$$\boldsymbol{C}\boldsymbol{S}_p\boldsymbol{C}' = \begin{pmatrix} 27.900 & -8.575 & 14.400 & -4.425 \\ -8.575 & 19.950 & -16.375 & -5.750 \\ 14.400 & -16.375 & 20.050 & 5.250 \\ -4.425 & -5.750 & 5.250 & 12.700 \end{pmatrix}$$

$$\left(\boldsymbol{C}\boldsymbol{S}_p\boldsymbol{C}'\right)^{-1} = \begin{pmatrix} 0.0918 & -0.0310 & -0.1076 & 0.0625 \\ -0.0310 & 0.1667 & 0.1589 & -0.0017 \\ -0.1076 & 0.1589 & 0.2782 & -0.0812 \\ 0.0625 & -0.0017 & -0.0812 & 0.1334 \end{pmatrix}$$

$$T^2 = \frac{n_1 n_2}{n_1+n_2}\left(\boldsymbol{C}(\bar{\boldsymbol{x}}_B - \bar{\boldsymbol{x}}_A)\right)'\left(\boldsymbol{C}\boldsymbol{S}_p\boldsymbol{C}'\right)^{-1}\boldsymbol{C}\left(\bar{\boldsymbol{x}}_B - \bar{\boldsymbol{x}}_A\right)$$

Let $\boldsymbol{y} = \left(CS_pC'\right)^{-1}C(\bar{\boldsymbol{x}}_B - \bar{\boldsymbol{x}}_A) = (0.8136, -1.2855, -1.8028, 0.2628)'$,

Then $T^2 = \frac{n_1 n_2}{n_1+n_2}\left(C(\bar{\boldsymbol{x}}_B - \bar{\boldsymbol{x}}_A)\right)'\boldsymbol{y} = 14.2582$. So,

$$T^2 = \frac{5\times 5}{5+5}\times 14.2582 = 35.645$$

$$T_\alpha^2 = \frac{k\left(n_1+n_2-2\right)}{n_1+n_2-k-1}F_{0.05}\left(k, n_1+n_2-k-1\right)$$

$$= \frac{4\times\left(5+5-2\right)}{5+5-4-1}F_{0.05}\left(4,5\right) = 6.4\times 5.19 = 33.216$$

Since $T^2 = 35.645 > 33.216 = T_\alpha^2$, we reject H_0 when α=0.05 (p=0.044).

This implies there exists a significant difference in the indices between Brand A and B.

Assignment：See the assignment in BB

§ 4.6 Multiple Population Mean Comparative Test （Multivariate Analysis of Variance）

- Suppose k population $\pi_1,\pi_2,\cdots,\pi_k$ and their distributions are respectively $N_p(\boldsymbol{\mu}_1,\boldsymbol{\Sigma}),N_p(\boldsymbol{\mu}_2,\boldsymbol{\Sigma}),\cdots,N_p(\boldsymbol{\mu}_k,\boldsymbol{\Sigma})$. Choose one sample independently from each of k populations, $\boldsymbol{x}_{i1},\boldsymbol{x}_{i2},\cdots,\boldsymbol{x}_{in_i}$, $i=1,2,\cdots,k$. Hypothesis test is

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_k, H_1: \mu_i \neq \mu_j, \text{ at least one couple } i \neq j \quad (4.5.1)$$

Denote by the following matrices

Total sum of squares $$\boldsymbol{SST} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(\boldsymbol{x}_{ij}-\overline{\boldsymbol{x}}\right)\left(\boldsymbol{x}_{ij}-\overline{\boldsymbol{x}}\right)' \quad (4.5.2)$$

Error sum of squares $$\boldsymbol{SSE} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(\boldsymbol{x}_{ij}-\overline{\boldsymbol{x}}_i\right)\left(\boldsymbol{x}_{ij}-\overline{\boldsymbol{x}}_i\right)' \quad (4.5.3)$$

Total regression sum of squares $$\boldsymbol{SSTR} = \sum_{i=1}^{k} n_i\left(\overline{\boldsymbol{x}}_i-\overline{\boldsymbol{x}}\right)\left(\overline{\boldsymbol{x}}_i-\overline{\boldsymbol{x}}\right)' \quad (4.5.4)$$

- Denote

$$\boldsymbol{SST} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(\boldsymbol{x}_{ij} - \bar{\boldsymbol{x}}\right)\left(\boldsymbol{x}_{ij} - \bar{\boldsymbol{x}}\right)' \longrightarrow df = n - 1$$

$$\boldsymbol{SSE} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(\boldsymbol{x}_{ij} - \bar{\boldsymbol{x}}_i\right)\left(\boldsymbol{x}_{ij} - \bar{\boldsymbol{x}}_i\right)' \longrightarrow E \longrightarrow df = n - k$$

$$\boldsymbol{SSTR} = \sum_{i=1}^{k} n_i\left(\bar{\boldsymbol{x}}_i - \bar{\boldsymbol{x}}\right)\left(\bar{\boldsymbol{x}}_i - \bar{\boldsymbol{x}}\right)' \longrightarrow H \longrightarrow df = k - 1$$

Then

$$\boldsymbol{SST} = \boldsymbol{SSE} + \boldsymbol{SSTR} \tag{4.5.5}$$

SST、***SSE*** and ***SSTR*** are sum of squares of total deviations and cross product matrix (总偏差平方和及叉积矩阵)、 sum of error squares and cross product matrix (误差(或组内) 平方和及叉积矩阵(简称组内矩阵)) and sum of regression squares and cross product matrix (处理平方(或组间)和及叉积矩阵(简称组间矩阵)), respectively, with free degree $(n-1)$, $(n-k)$ and $(k-1)$ separately.

Using likelihood ratio method, we can obtain Wilks Λ Statistic（威尔克斯Λ统计量）

$$\Lambda_{p,k-1,n-k} = \frac{|SSE|}{|SSE+SSTR|} = \frac{|SSE|}{|SST|} \quad (4.5.6)$$

Proof: It is noticed that the likelihood function is

$$\mathrm{L}(\mu,\Sigma)=\prod_{i=1}^{k}(2\pi)^{-n_i p/2}|\Sigma|^{-n_i/2}\exp\{-\tfrac{1}{2}\textstyle\sum_{j=1}^{n_i}(x_{ij}-\mu_i)'\Sigma^{-1}(x_{ij}-\mu_i)\}$$

$$= (2\pi)^{-np/2}|\Sigma|^{-\sum_{i=1}^{k}\frac{n_i}{2}}\exp\{-\tfrac{n}{2}\textstyle\sum_{i=1}^{k} trace[\Sigma^{-1}\underbrace{\tfrac{1}{n}\textstyle\sum_{j=1}^{n_i}(x_{ij}-\mu_i)(x_{ij}-\mu_i)'}_{\text{An approximation of }\Sigma}]\}$$

$$\approx (2\pi)^{-np/2}|\Sigma|^{-\frac{n}{2}}\exp\left(-\frac{np}{2}\right)$$

with $\mathrm{n} = \sum_{i=1}^{k} n_i$. Let $\hat{\Sigma}_0$ and $\hat{\Sigma}$ be the MLEs of Σ under H_0 and H_1. Then, $\hat{\Sigma}_0 = \frac{1}{n}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij}-\bar{x})(x_{ij}-\bar{x})'$ and $\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij}-\bar{x}_i)(x_{ij}-\bar{x}_i)'$. Therefore, the likelihood ratio statistic is defined as

$$\lambda = \frac{max_{\mu_1=\cdots=\mu_k,\Sigma}L(\mu,\Sigma)}{max_{\mu,\Sigma}L(\mu,\Sigma)} = \frac{|\hat{\Sigma}_0|^{-n/2}}{|\hat{\Sigma}|^{-n/2}} = \frac{|\frac{1}{n}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij}-\bar{x})(x_{ij}-\bar{x})'|^{-n/2}}{|\frac{1}{n}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij}-\bar{x}_i)(x_{ij}-\bar{x}_i)'|^{-n/2}} = \frac{|\mathrm{SST}|^{-n/2}}{|\mathrm{SSE}|^{-n/2}}.$$

This leads to $\frac{2}{n}\ln(\lambda) = ln\frac{|SSE|}{|SST|}$. Then, we define Wilks Λ Statistic as

$$\Lambda \equiv \exp\left\{\frac{2}{n}\ln(\lambda)\right\} = \frac{|\mathrm{SSE}|}{|\mathrm{SST}|},$$

which ranges between 0 and 1.

Using likelihood ratio method, we can obtain Wilks Λ Statistic （威尔克斯Λ统计量）

$$\Lambda_{p,k-1,n-k} = \frac{|SSE|}{|SSE+SSTR|} = \frac{|SSE|}{|SST|} \qquad (4.5.6)$$

with $\Lambda_{p,k-1,n-k} \in [0,1]$.

For a given significance level α, the rejection rule is : H_0

$$\text{If } \Lambda_{p,k-1,n-k} \le \Lambda_{p,k-1,n-k,\alpha}, \text{reject } H_0 \qquad (4.5.7)$$

where the critical value $\Lambda_{p,k-1,n-k,\alpha}$ satisfies: when H_0 is true,

$$P(\Lambda_{p,k-1,n-k} \le \Lambda_{p,k-1,n-k,\alpha}) = \alpha \qquad (4.5.8)$$

$\Lambda_{p,m,r,\alpha}$ are usually obtained from F distribution (or Chi-square distribution) table.

$$\frac{(r-p+1)(1-\sqrt{\Lambda_{p,2,r}})}{p\sqrt{\Lambda_{p,2,r}}} \sim F(2p, 2(r-p+1))$$

Property : when m=2, $r = n - k$.

Example 4.6.1 In order to study the effect of sales modes to commodity sale, we elect 4 kinds of goods (A、B、C and D) and each kind is sold by means of 3 sales method(I、 II and III). Suppose commodity sale of goods are x_1, x_2, x_3, x_4 . Data are listed in Table 4.6.1。

Table 4.6.1 Sale Data

Number	I				II				III			
	x_1	x_2	x_3	x_4	x_1	x_2	x_3	x_4	x_1	x_2	x_3	x_4
1	125	60	338	210	66	54	455	310	65	33	480	260
2	119	80	233	330	82	45	403	210	100	34	468	295
3	63	51	260	203	65	65	312	280	65	63	416	265
4	65	51	429	150	40	51	477	280	117	48	468	250
5	130	65	403	205	67	54	481	293	114	63	395	380
6	69	45	350	190	38	50	468	210	55	30	546	235
7	46	60	585	200	42	45	351	190	64	51	507	320

8	146	66	273	250	113	40	390	310	110	90	442	225
9	87	54	585	240	80	55	520	200	60	62	440	248
10	110	77	507	270	76	60	507	189	110	69	377	260
11	107	60	364	200	94	33	260	280	88	78	299	360
12	130	61	391	200	60	51	429	190	73	63	390	320
13	80	45	429	270	55	40	390	295	114	55	494	240
14	60	50	442	190	65	48	481	177	103	54	416	310
15	81	54	260	280	69	48	442	225	100	33	273	312
16	135	87	507	260	125	63	312	270	140	61	312	345
17	57	48	400	285	120	56	416	280	80	36	286	250
18	75	52	520	260	70	45	468	370	135	54	468	345
19	76	65	403	250	62	66	416	224	130	69	325	360
20	55	42	411	170	69	60	377	280	60	57	273	260

- In this case, the hypothesis test problem is

$H_0: \ \boldsymbol{\mu}_1=\boldsymbol{\mu}_2=\boldsymbol{\mu}_3, \quad H_1: \ \boldsymbol{\mu}_1,\boldsymbol{\mu}_2,\boldsymbol{\mu}_3$ at least one couple are not equal

where $\boldsymbol{\mu}_1,\boldsymbol{\mu}_2,\boldsymbol{\mu}_3$ are respectively population mean vectors of sale method I, II and III. Assume all populations are multivariate normal and have the same covariance matrix

$$p=4, \quad k=3, \quad n_1=n_2=n_3=20, \quad n=n_1+n_2+n_3=60$$

$$\overline{\boldsymbol{x}}_1=\begin{pmatrix}90.80\\58.65\\404.50\\230.65\end{pmatrix}, \quad \overline{\boldsymbol{x}}_2=\begin{pmatrix}72.90\\51.45\\417.75\\253.15\end{pmatrix}, \quad \overline{\boldsymbol{x}}_3=\begin{pmatrix}94.15\\55.15\\403.75\\292.00\end{pmatrix}$$

$$\overline{\boldsymbol{x}}=\frac{1}{n}\sum_{i=1}^{3}n_i\overline{\boldsymbol{x}}_i=\frac{1}{3}\sum_{i=1}^{3}\overline{\boldsymbol{x}}_i=\begin{pmatrix}85.9500\\55.0833\\408.6667\\258.6000\end{pmatrix}$$

$$\boldsymbol{SSTR} = \sum_{i=1}^{3} n_i \overline{\boldsymbol{x}}_i \overline{\boldsymbol{x}}_i' - n\overline{\boldsymbol{x}}\overline{\boldsymbol{x}}' = \begin{pmatrix} 5221.30 & 1305.20 & -3581.25 & 4188.90 \\ 1305.20 & 518.53 & -963.83 & -1553.20 \\ -3581.25 & -963.83 & 2480.83 & -1945.25 \\ 4188.90 & -1553.20 & -1945.25 & 38529.30 \end{pmatrix}$$

$$\boldsymbol{SST} = \sum_{i=1}^{3} \sum_{j=1}^{n_i} \boldsymbol{x}_{ij} \boldsymbol{x}_{ij}' - n\overline{\boldsymbol{x}}\overline{\boldsymbol{x}}' = \begin{pmatrix} 49290.85 & 8992.25 & -36444.00 & 28906.80 \\ 8992.25 & 9666.58 & -4658.33 & 4859.00 \\ -36444.00 & -4658.33 & 429509.33 & -58114.00 \\ 28906.80 & 4859.00 & -58114.00 & 175644.40 \end{pmatrix}$$

$$\boldsymbol{SSE} = \boldsymbol{SST} - \boldsymbol{SSTR} = \begin{pmatrix} 44069.55 & 7687.05 & -32862.75 & 24717.90 \\ 7687.05 & 9148.05 & -3694.50 & 6412.20 \\ -32862.75 & -3694.50 & 427028.50 & -56168.75 \\ 24717.90 & 6412.20 & -56168.75 & 137115.10 \end{pmatrix}$$

Therefore

$$\Lambda_{4,2,57}=\frac{|\boldsymbol{SSE}|}{|\boldsymbol{SST}|}=\frac{1.6464\times10^{19}}{2.4708\times10^{19}}=0.6663$$

According to equation(4−3.4) in Appendix 4−3:

$$F=\frac{(57-4+1)\left(1-\sqrt{0.6663}\right)}{4\times\sqrt{0.6663}}=3.039 \qquad \frac{(r-p+1)(1-\sqrt{\Lambda_{p,2,r}})}{p\sqrt{\Lambda_{p,2,r}}}\sim F(2p,2(r-p+1))$$

From F-Table, $F_{0.01}(8,108)$=2.68 $<$ 3.039= F. Then reject H_0 when α=0.01. Therefore, we can draw the conclusion that sale amount of 3 sale methods are significantly different(p=0.004).

- In order to find out which commodity maybe cause the significant difference of the three sales modes, we utilize univariate variance analysis to test the 4 kinds of goods. Use diagonal elements of ***SSTR*** and ***SSE:*** sales modes to commodity sale

$$F_1 = \frac{5221.30/2}{44069.55/57} = 3.377,$$

$$F_2 = \frac{518.53/2}{9148.05/57} = 1.615$$

$$F = \frac{SSTR/(k-1)}{SSE/(n-k)} \sim F(k-1, n-k)$$

$$F_3 = \frac{2480.83/2}{427028.50/57} = 0.166,$$

$$F_4 = \frac{38529.30/2}{137115.10/57} = 8.008$$

Apply P-value rule to make a judge :

From F-Table, $F_{0.05}(2,57)$=3.16, $F_{0.01}(2,57)$=5.01. So, product A is significantly different(p=0.041). Product D is extremely significantly different(p=0.001). However, product B and C are insignificantly different(p=0.208 and p=0.848).

- If we get rid of product D and test 3 other kinds of goods through Λ statistic, then

$$\Lambda_{3,2,57} = \frac{|\boldsymbol{SSE}|}{|\boldsymbol{SST}|} = \frac{1.3831\times 10^{14}}{1.5906\times 10^{14}} = 0.8695$$

$$\frac{(r-p+1)(1-\sqrt{\Lambda_{p,2,r}})}{p\sqrt{\Lambda_{p,2,r}}} \sim F(2p,2(r-p+1))$$

$$F = \frac{(57-3+1)\left(1-\sqrt{0.8695}\right)}{3\times\sqrt{0.8695}} = 1.328$$

$F_{0.05}(6,110)=2.18>1.328$, insignificant. Therefore, for product A、B and C, population mean vectors of sale method I、II and III are insignificantly different(p=0.251).

§ 4.7 The covariance matrix equality Test

- Suppose k population $\pi_1,\pi_2,\cdots,\pi_k$ and their distributions are respectively $N_p(\boldsymbol{\mu}_1,\boldsymbol{\Sigma}),N_p(\boldsymbol{\mu}_2,\boldsymbol{\Sigma}),\cdots,N_p(\boldsymbol{\mu}_k,\boldsymbol{\Sigma})$. Choose one sample independently from each of k populations, $\boldsymbol{x}_{i1},\boldsymbol{x}_{i2},\cdots,\boldsymbol{x}_{in_i}$, i=1,2,$\cdots$,k. Hypothesis test is

$$H_0:\ \ \boldsymbol{\Sigma}_1=\boldsymbol{\Sigma}_2=\cdots=\boldsymbol{\Sigma}_k,\quad H_1:\ \ \boldsymbol{\Sigma}_i\neq\boldsymbol{\Sigma}_j,\text{ at least one couple } i\neq j \quad (4.7.1)$$

Let $n=\sum_{i=1}^{k}n_i$. The likelihood function of the sample $\boldsymbol{x}_{ij},j=1,\cdots,n_i$,

is express as $x_{i1},x_{i2},\cdots,x_{in_i},i=1,2,\cdots,k$

$$L(\mu_1,\Sigma_1,\cdots,\mu_k,\cdots,\Sigma_k)=\prod_{i=1}^{k}L_i(\mu_i,\Sigma_i).$$

The likelihood ratio statistic is formulated as

$$\lambda=\frac{\max\limits_{\mu_i,\Sigma>0}L(\mu_1,\cdots,\mu_k,\Sigma)}{\max\limits_{\mu_i,\Sigma_i>0}L(\mu_1,\Sigma_1,\cdots,\mu_k,\cdots,\Sigma_k)}$$

§ 4.7 The covariance matrix equality Test

It can be proved that (Exercise 1) likelihood ratio statistic

$$\lambda = \frac{\left|\frac{A}{n}\right|^{-n/2}}{\prod_{i=1}^{k}\left|\frac{A_i}{n_i}\right|^{-n_i/2}},$$

where $A_i = \sum_{j=1}^{n_i}(x_{ij}-\bar{x}_i)'(x_{ij}-\bar{x}_i)$ and $A = \sum_{i=1}^{n}\sum_{j=1}^{n_i}(x_{ij}-\bar{x}_i)'(x_{ij}-\bar{x}_i)$ such that $A = A_1 + \cdots + A_k$.

A modification of the above LRT statistic based on unbiasedness is

$$\lambda = \frac{|S_p|^{(n-k)/2}}{\prod_{i=1}^{k}|S_i|^{(n_i-1)/2}}, \qquad (4.7.2)$$

where $S_i = \frac{1}{n_i-1}\sum_{j=1}^{n_i}(x_{ij}-\bar{x}_i)(x_{ij}-\bar{x}_i)'$, $\bar{x}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} x_{ij}$, and $S_p = \frac{1}{n-k}\sum_{i=1}^{k}(n_i-1)S_i$.

Then, Box' M statistic is defined as:

$$M = -2\ln\lambda = (n-k)\ln|S_p| - \sum_{i=1}^{k}(n_i-1)\ln|S_i|. \qquad (4.7.3)$$

Proofs of (4.7.3) can be founded in Chapt 4.7 Box-M test (see BB).

§ 4.7 The covariance matrix equality Test

A modified LRT statistic for the hypothesis test (4.7.1) is

$$\lambda = \frac{|S_p|^{(n-k)/2}}{\prod_{i=1}^{k} |S_i|^{(n_i-1)/2}}, \qquad (4.7.2)$$

where $S_i = \frac{1}{n_i-1}\sum_{j=1}^{n_i}(x_{ij}-\overline{x_i})(x_{ij}-\overline{x_i})'$, $\overline{x_i} = \frac{1}{n_i}\sum_{j=1}^{n_i} x_{ij}$, and $S_p = \frac{1}{n-k}\sum_{i=1}^{k}(n_i-1)S_i$.

Then, Box' M statistic is defined as:

$$M = -2\ln\lambda = (n-k)\ln|S_p| - \sum_{i=1}^{k}(n_i-1)\ln|S_i|. \qquad (4.7.3)$$

Proofs of (4.7.3) can be founded in Chapt 4.7 Box-M test (see BB).

Under H_0, $u = (1-c)M$ is approximately distributed as a chi-square distribution with degree of $\frac{1}{2}(k-1)p(p+1)$ [explanation can be found in next page], where

$$c = \left(\sum_{i=1}^{k}\frac{1}{n_i-1} - \frac{1}{n-k}\right)\frac{2p^2+2p-1}{6(p+1)(k-1)} \text{ and } c = \frac{(2p^2+3p-1)(k+1)}{6(p+1)(n-k)}$$

if all n_i are equal. For a given level of significance α, the reject rule is

$$\text{if } u \geq \chi^2_\alpha[\tfrac{1}{2}(k-1)p(p+1)], \text{ then reject } H_0.$$

§ 4.7 The covariance matrix equality Test

Box' M statistic:

$$M=-2\ln\lambda=(n-k)\ln\left|S_p\right|-\sum_{i=1}^{k}(n_i-1)\ln|S_i|. \qquad (4.7.3)$$

Under H_0 is true, $\mathrm{u}=(1-\mathrm{c})M$ is approximately distributed as a chi-square distribution with degree of $\frac{1}{2}(k-1)p(p+1)$, where

$$c=\left(\sum_{i=1}^{k}\frac{1}{n_i-1}-\frac{1}{n-k}\right)\frac{2p^2+2p-1}{6(p+1)(k-1)} \text{ and } c=\frac{(2p^2+3p-1)(k+1)}{6(p+1)(n-k)} \text{ if all } n_i \text{ are equal.}$$

参考 Box (1949)，有如下定理

> 当样本容量 n 很大时，
>
> $$-2\ln\lambda=-2\ln\left(\frac{\max_{\theta\in\Theta_0}L(X;\theta)}{\max_{\theta\in\Theta}L(X;\theta)}\right)$$
>
> 近似服从自由度为 f 的 χ^2 分布，其中 $f=\dim\Theta-\dim\Theta_0$

所以有

$$\begin{aligned} f&=\dim\Theta-\dim\Theta_0\\ &=\sum_{i=1}^{k}f_{\Sigma_i}-f_{\Sigma}\\ &=k\cdot\frac{p(p+1)}{2}-\frac{p(p+1)}{2}\\ &=\frac{(k-1)p(p+1)}{2}\end{aligned}$$

§ 4.7 The covariance matrix equality Test

Example 4.7.1: In example 4.6.1, we hope to test

$$H_0\colon\ \Sigma_1 = \Sigma_2 = \Sigma_3,\ H_1\colon \text{ at least two of } \Sigma_1, \Sigma_2, \Sigma_3 \text{ are not equal.}$$

Solution: by calculation, we have

$$|S_1| = 1.004 \times 10^{12},\ |S_2| = 4.828 \times 10^{11}$$

$$|S_3| = 2.0339 \times 10^{12},\ \left|S_p\right| = 1.5597 \times 10^{12}.$$

Take the natural logarithm, we obtain

$$ln|S_1| = 27.6358,\ ln|S_2| = 26.9030,\ ln|S_3| = 28.3410,\ ln\left|S_p\right| = 28.0755.$$

Further, we have

$$M = (60-3) \times 28.0755 - (20-1)(27.6358 + 26.9.30 + 28.3410) = 25.5873,$$

$$u = (1-c)M = \left[1 - \frac{(2\times4^2+3\times4-1)(3+1)}{6\times(4+1)(60-3)}\right] \times 25.5873 = 23.014. \qquad \mathrm{c} = \frac{(2p^2+3p-1)(k+1)}{6(p+1)(n-k)}$$

The degree of freedom is $0.5 \times$ (3-1) $\times 4 \times (4+1) = 20$. From the table of chi-square distribution, we have $\chi^2_{0.05}(20) = 31.410 > 23.014 = u$.

Therefore, we do not reject H_0 and think there is no significant difference in the covariance matrix of the three sales methods

Exercise 2 (Choice question). Wilks Λ statistics is often used in the comparative test of K normal population means. The definition of Wilks Λ statistics is as follows: $\Lambda_{p,k-1,n-k} = |SSE|/|SST|$, where n is the total sample size from k normal populations, SSE is the error sum of squares, SST is the sum of squares of total deviation. Under which of the following circumstances will the statistic result in rejection of the original hypothesis (B)

A larger B smaller C invariant D no judgement

Exercise 3. We want to study whether the new airport is different from the old airport in terms of the pollutions caused by airplane traffic. The pollutions are measured by two standards: air pollution and noise pollution. For simplicity, we consider the following data of three random samples collected from the old airport and three random samples collected from the new airport:

$$\text{The old airport}: \quad X_{11} = \begin{pmatrix} 10 \\ 2 \end{pmatrix} \quad X_{12} = \begin{pmatrix} 9 \\ 3 \end{pmatrix} \quad X_{13} = \begin{pmatrix} 11 \\ 2 \end{pmatrix};$$

$$\text{The new airport}: \quad X_{21} = \begin{pmatrix} 8 \\ 3 \end{pmatrix} \quad X_{22} = \begin{pmatrix} 9 \\ 4 \end{pmatrix} \quad X_{23} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}.$$

Here the first component is air pollution index and the second is noise pollution index. Based on this data, please develop the corresponding test procedures to answer this question? (Assume the variances of the sampled pollution are same for the two airports and the populations are from bivariate normal distribution.)

Assignment： See the assignment file in BB